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EXAM PREP

Digital for ISRO

Expected Questions from Digital

By Sanjay Sir

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- Teaching Experience: 21+ Years
- Achievements:
  - AIR-16 (ESE 2005), AIR-21 (ESE 2008)
  - Mentored More Than 1,00,000+ Students
- Area of Expertise: Network Theory, Digital Circuits, Control Systems, EDC, Analog Circuits, Advanced Electronics

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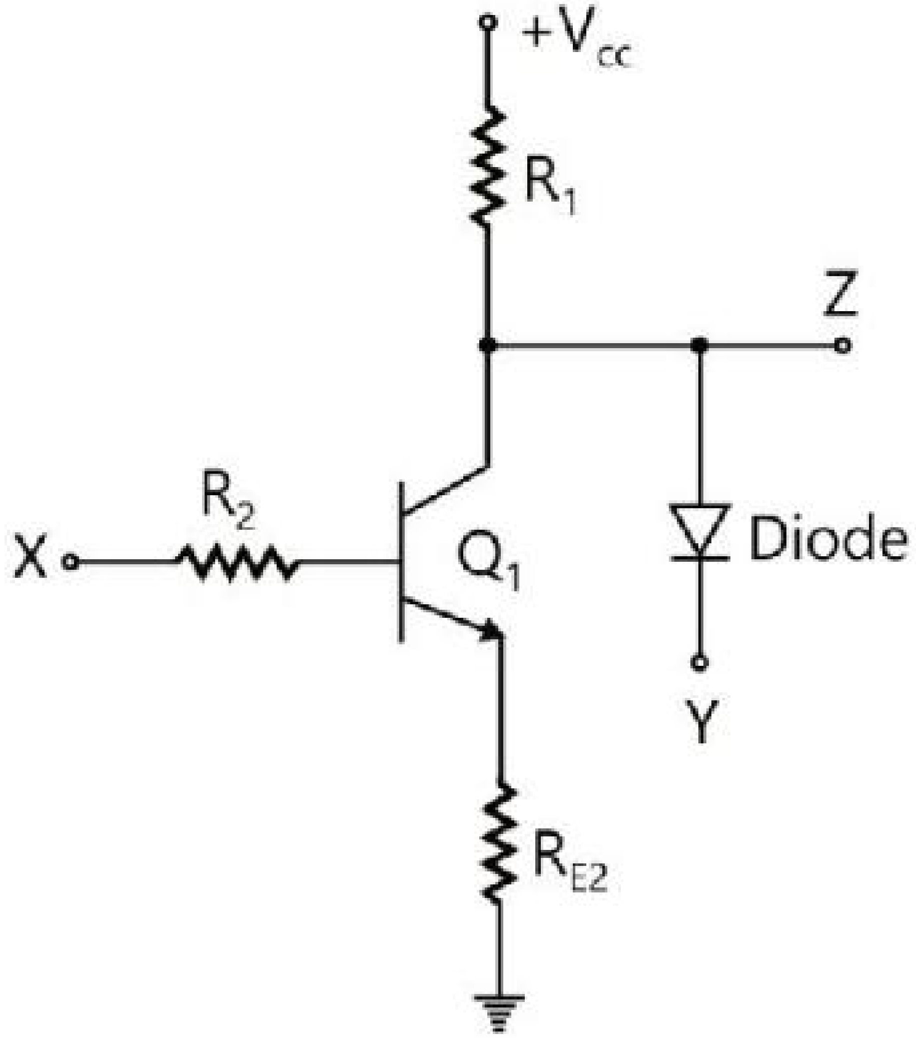


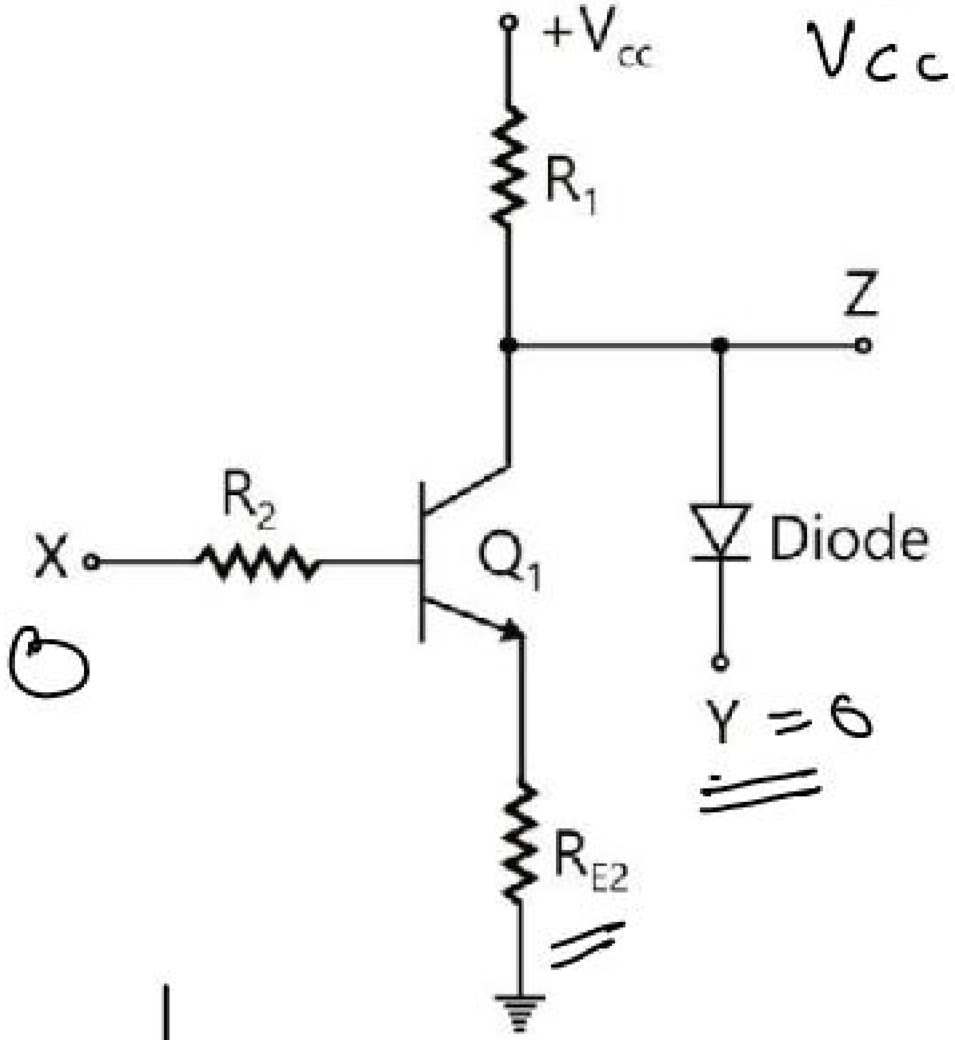
Q. In the circuit shown, Q_1 has negligible collector-to-emitter saturation voltage and the diode drops negligible voltage across it under forward bias. If V_{cc} is +5 V, X and Y are digital signals with 0 V as logic 0 and V_{cc} as logic 1, the Boolean expression for Z is

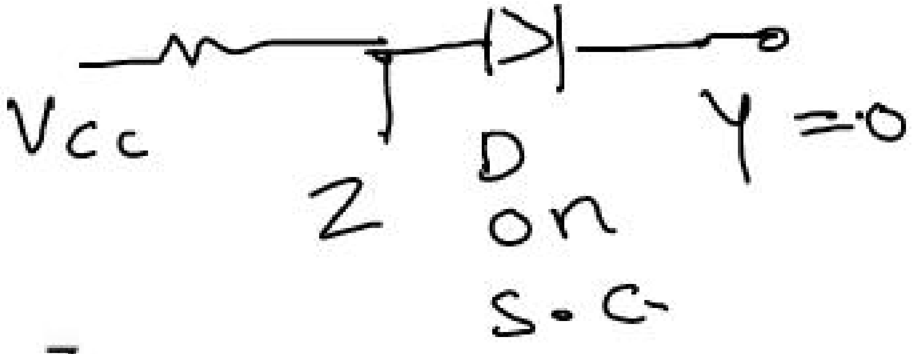
A. XY

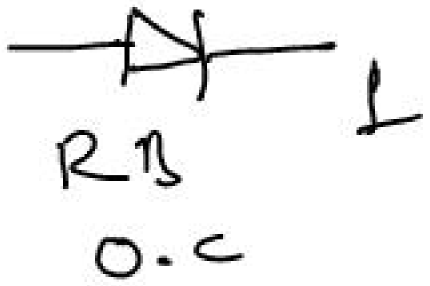
B. X'Y Ans.

C. XY'

D. (XY)'

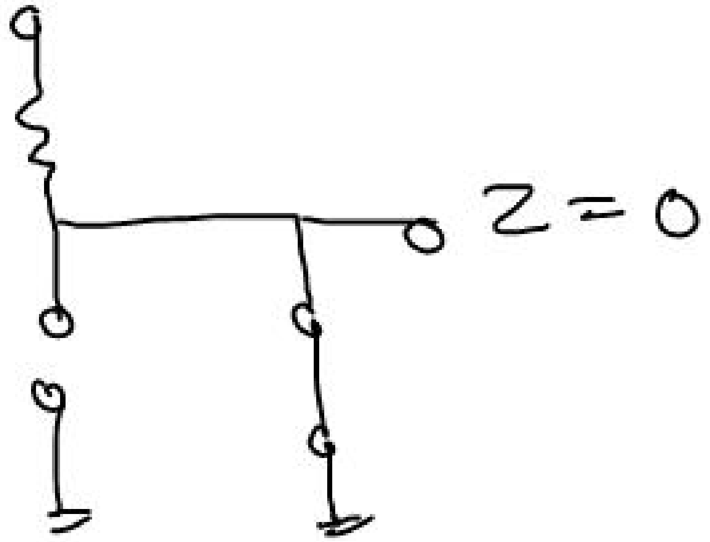


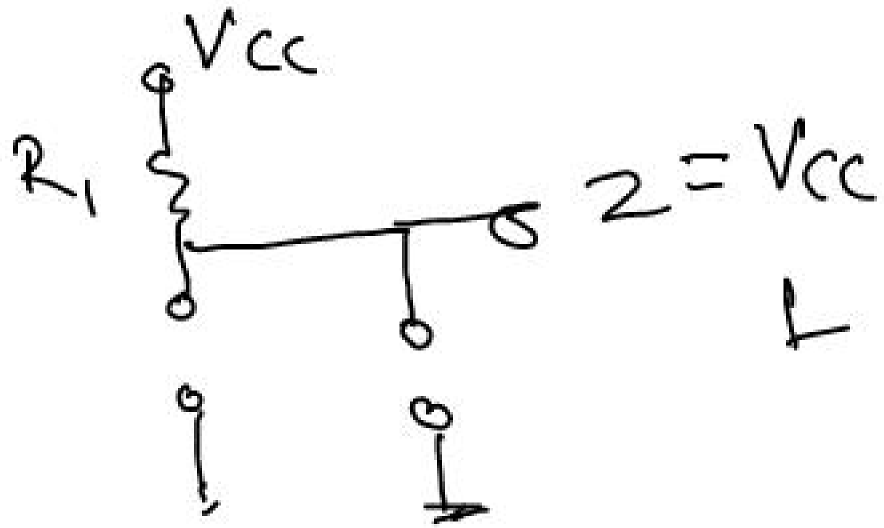


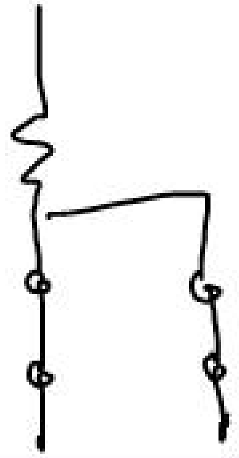


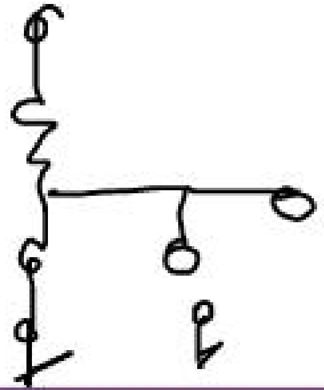
X Y	Q_1 D	Z
0 0	off on	0
0 1	off off	1
1 0	on on	0
1 1	on off	0

$$\boxed{Z = \overline{X}Y}$$









Q. Which one of the following is the correct answer when 11011_2 is subtracted from 11101_2 by using the 1's complement method?

A. 01001

B. 10001

C. 00011

D. 00010

A

B<A

Analog

11:00 AM

$$\begin{array}{r} 11101 \\ -\ 11011 \\ \hline \end{array}$$

$$\begin{array}{r} 11101 \\ +\ 00100 \\ \hline \boxed{1}\ 00001 \\ 1 \\ \hline 00010 \end{array}$$

$$\begin{array}{r} 11101 \\ 11011 \\ \hline 00010 \end{array}$$

Quiz → N/w

Quiz → Analog.

Q. A number in 4-bit two's complement representation is $X_3X_2X_1X_0$. This number when stored using 8-bits will be

A. 0000 $X_3X_2X_1X_0$

B. 1111 $X_3X_2X_1X_0$

C. $X_3X_3X_3X_3$ $X_3X_2X_1X_0$

D. $X_3'X_3'X_3'X_3'$ $X_3X_2X_1X_0$

X_3

3 0011 +3

0000 0011 +3

−3 0011 −3

−3 1101

0000 0011

1111 1101

4 bit $a_3 a_2 a_1 a_0$ N

$a_3 a_3 a_3 a_2 a_1 a_0$ N

$a_3 a_3 a_2 a_1 a_0 0$

Q. A number N is stored in a 4-bit 2's complement representation as

a_3	a_2	a_1	a_0

It is copied into a 6-bit register and after a few operations, the final bit pattern is

a_3	a_3	a_2	a_1	a_0	1

The value of this bit pattern in 2's complement representation is given in terms of the original number N as

A. $32a_3+2N+1$
B. $32a_3-2N-1$
C. $2N-1$
D. $2N+1$

0011 N

3 000011 N

6 000110 2N

$+$ 1

000111 2N+1

0100 4

1000 8

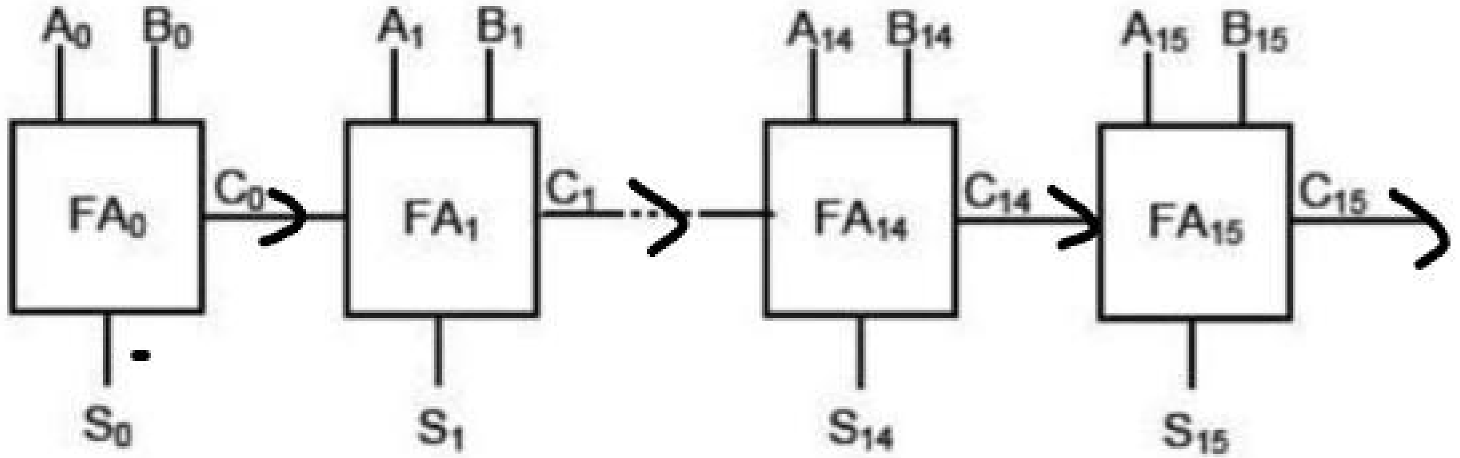
10000 16

100000 32

FA HA FS HS

MCQ.

Q. A 16-bit ripple carry adder is realized using 16 identical full adders (FA) as shown in the figure. The carry propagation delay of each FA is 12 ns and the sum-propagation delay of each FA is 15 ns. The worst case delay (in ns) of this 16-bit adder will be ______.



$$\boxed{\tau_{delay} = (n-1)\tau_c + \max[\tau_c, \tau_s]}$$

$$= (16-1)\times 12 + \max[12, 15]$$

$$= 15\times 12 + 15$$

$$= 180 + 15$$

$$= \underline{\underline{195\ n\,sec}}$$

Q. If A and B are 4 bit Binary numbers given to a 4 bit comparator, then the number of combinations for which A > B is __________.

A. 120

B. 160

C. 116

D. 140

Magnitude comparator

$A = A_3 A_2 A_1 A_0$

$B = B_3 B_2 B_1 B_0$

$2^{2n} = 2^{2\times 4} = 256$

$(A=B) = 2^n = 2^4$

$(A>B) \text{ OR } (A<B) = \dfrac{2^{2n} - 2^n}{2} = \dfrac{2^8 - 2^4}{2} = \dfrac{240}{2}$

$= 120$

$2^{2n} = 2^{2\times 2} = 16$

A_1	A_0	B_1	B_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0

1101

1110

1111

A_3	A_2	A_1	A_0	B_3	B_2	B_1	B_0	
0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	255

$2^n = 2^4 = 16$

$(A>B)$ OR $(A<B)$

$A > B$

Expected Questions From Digital ISRO

	F
0000	0
0001	0
0010	0
0011	0
0100	0
0101	1
0110	1
0111	1
1000	1
1001	1
1010	X
1011	X
1100	X
1101	X
1110	X
1111	X

Q. A circuit outputs a digit in the form of 4 bits. 0 is represented by 0000, 1 by 0001,...9 by 1001. A combinational circuit is to be designed which takes these 4 bits as input and outputs 1 if the digit ≥ 5, and 0 otherwise. If only AND, OR and NOT gates may be used, what is the minimum number of gates required?

A. 2

B. 3

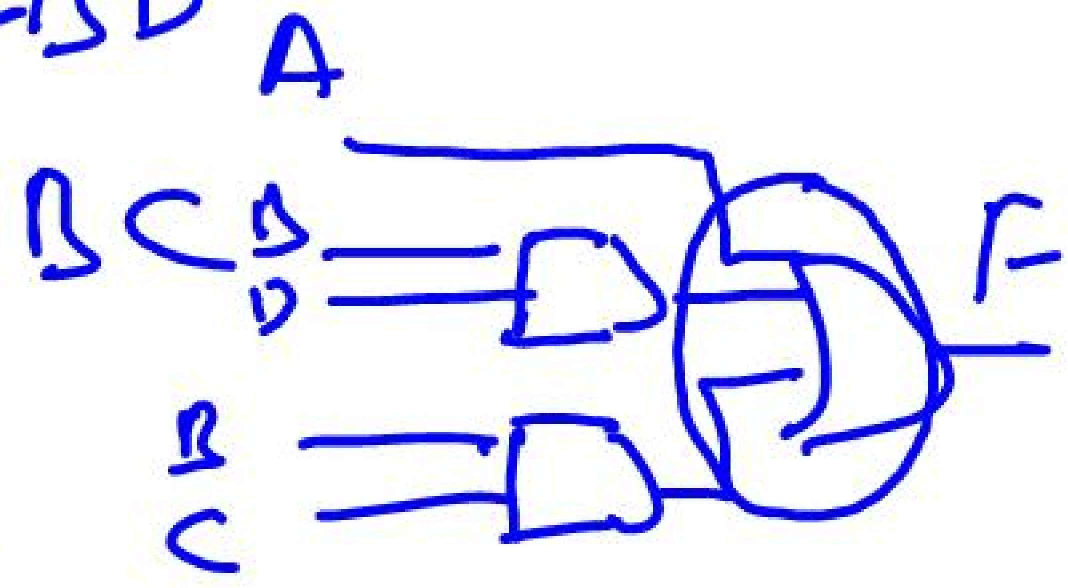
C. 4

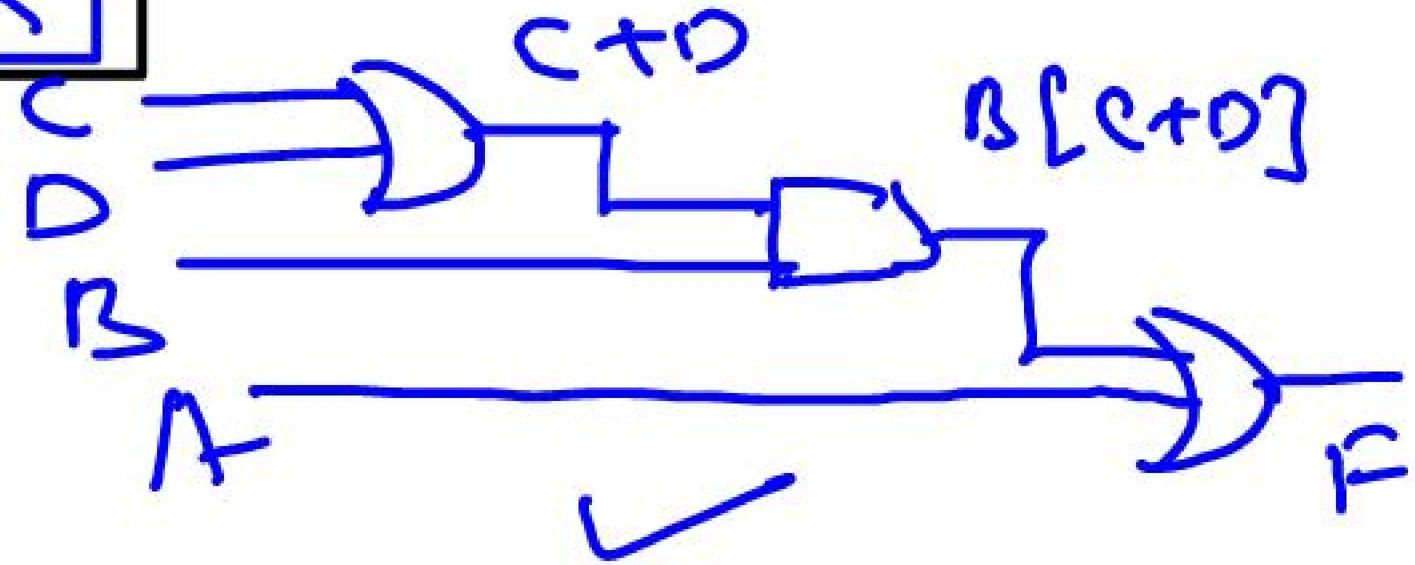
D. 5

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	3	2
$\bar{A}B$	4	5: 1	7: 1	6: 1
AB	12: X	13: X	15: X	14: X
$A\bar{B}$	8: 1	9: 1	11: X	10: X

BD

A

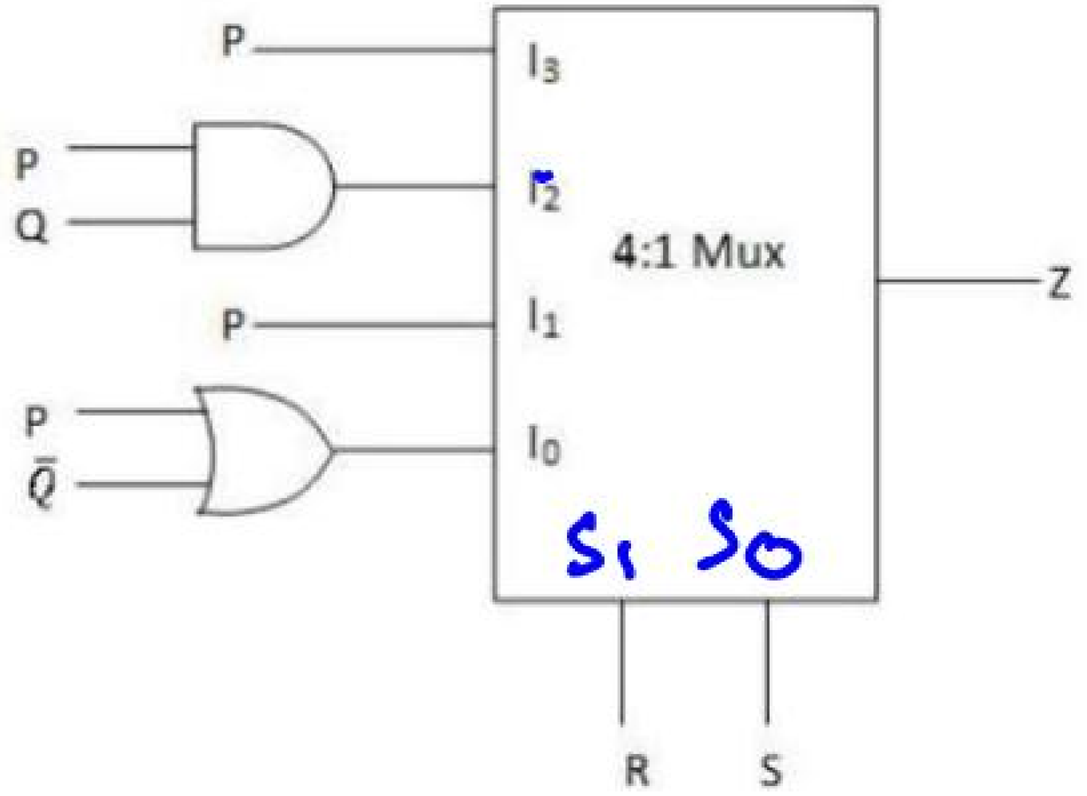




$A + BD + BC$

$A + B[C+D]$

Expected Questions From Digital ISRO



Q. For the circuit shown in the following figure, I_0 - I_3 are inputs to the 4 : 1 multiplexer. R(MSB) and S are control bits. The output Z can be represented by

A. PQ + PQ'S + Q'R'S'

B. PQ' + PQR' + P'Q'S'

C. PQ'R' + P'QR + PQRS + Q'R'S'

D. PQR' + PQRS' + PQ'R'S + Q'R'S'

$$F = \bar{S}_1\bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$

$$F = \bar{R}\bar{S}[P + \bar{Q}] + \bar{R} S P + R\bar{S} P Q + R S P$$

$$F = \bar{R}\bar{S}P + \bar{R}\bar{S}\bar{Q} + \bar{R}SP + R\bar{S}PQ + RSP$$

	$\bar{P}\bar{Q}$	$\bar{P}Q$	PQ	$P\bar{Q}$
$\bar{R}\bar{S}$	1		1	1
$\bar{R}S$			1	1
RS			1	1
$R\bar{S}$			1	

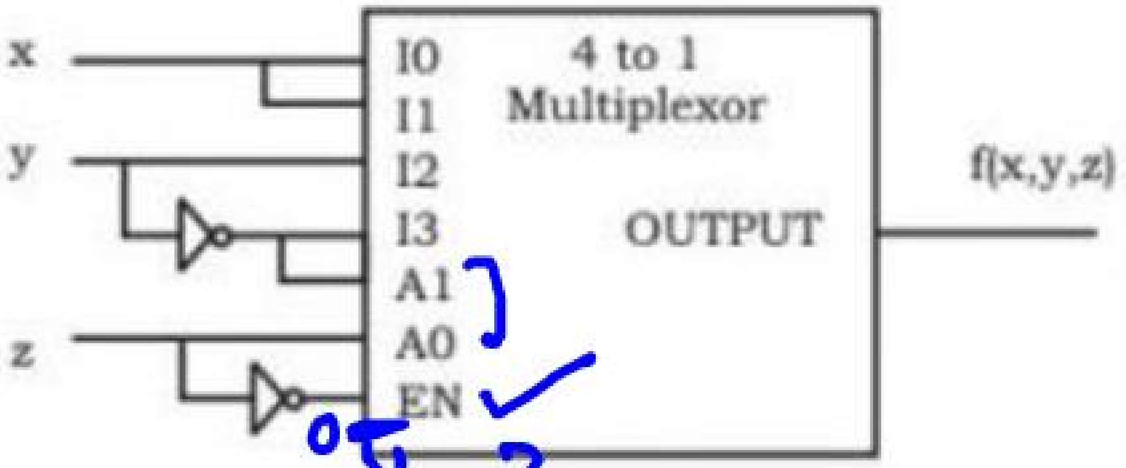
$$F = PQ + \bar{R}\bar{S}\bar{Q} + P\bar{Q}S$$

Q. Consider a multiplexer with X and Y as data inputs and Z as control input. Z = 0 selects input X, and Z = 1 selects input Y. What are the connections required to realize the 2-variable Boolean function f = T + R, without using any additional hardware ?

A. R to X, 1 to Y, T to Z
B. T to X, R to Y, T to Z
C. T to X, R to Y, 0 to Z
D. R to X, 0 to Y, T to Z

H.W.

[GATE CS -2014]



XYZ	$A_1 A_0$	$EN = \bar{z}$	Y
000	1 0	1	0
001		0	0
010	0 0	1	X=0
011		0	0
100	1 0		0 =Y
101		0	0
110	0 0		X=1
111		0	0

Q. Consider the following multiplexer where I0, I1, I2, I3 are four data input lines selected by two address line combinations A_1A_0 = 00, 01, 10, 11 respectively and f is the output of the multiplexer. EN is the enable input.

The function f(x, y, z) implemented by the above circuit is

A. xyz'
B. xy + z
C. x + z
D. None of these

$xy\bar{z}$

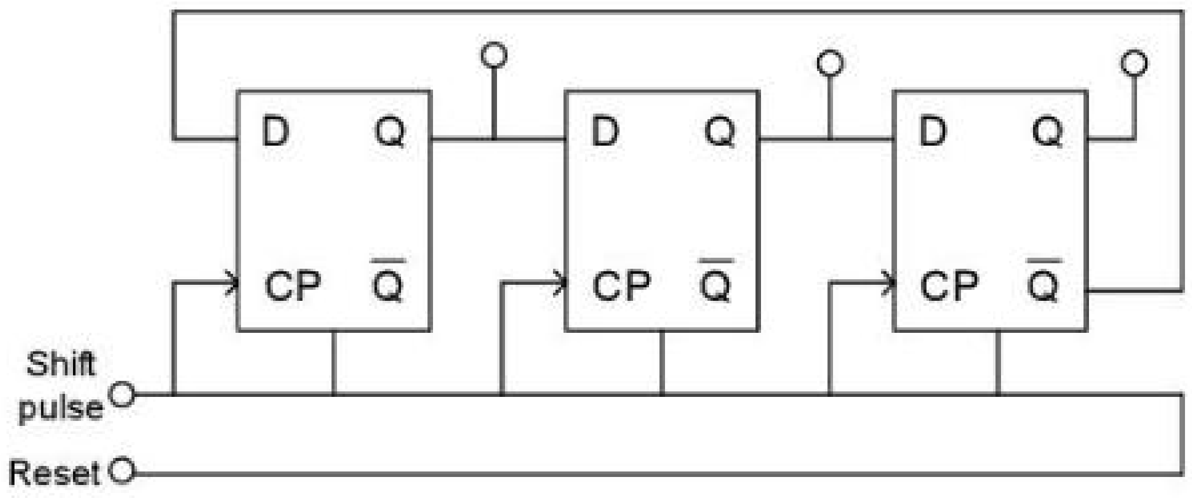
H.W.

Q. Consider the following statements regarding registers and latches:

1. Registers are made of edge - triggered FFs, whereas latches are made from level-triggered FFs.
2. Registers are temporary storage devices whereas latches are not.
3. A latch employs cross-coupled feedback connections.
4. A register stores a binary word whereas a latch does not.

Which of the statements given above are correct?

A. 1 and 2 B. 1 and 3 C. 2 and 3 D. 3 and 4



Q. A three – bit – shift register is shown in the given figure,

To have the content '000' again, the number of clock pulses required would be

A. 3

B. 6

C. 8

D. 16

H.W.

Q. Consider the following statements:

1.When two unsigned numbers are added, an overflow is detected from the carry into the most significant position.

2.An overflow does not occur if the two numbers added are both negative.

3.If the carry into the sign bit position and carry out of the sign bit position are not equal, an overflow condition is produced.

Which of the above statements is/are correct?

A. 1, 2 and 3

B. 1 only

C. 2 only

D. 3 only

H. W.

11:00 Am

Analog: